

Chapter 5 - The Foreign Exchange Market

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2026-03-01

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1 Introduction

Random Quote: “The old world is dying, and the new world struggles to be born: Now is the time of monsters.” - Antonio Gramsci

Another Random Quote: “Fairy tales do not tell children that dragons exist. Children already know this. Fairy tales tell children that dragons can be killed.” - G.K. Chesterton

Scripture: “You will hear of wars and rumors of wars, but see that you are not alarmed. Such things must happen, but the end is still to come.” Matthew 24:6

And: “I have told you these things, so that in me you may have peace. In this world you will have trouble. But take heart! I have overcome the world.” John 16:33

“The best way to destroy the capitalist system is to debauch the currency. By a continuing process of inflation, governments can confiscate, secretly and unobserved, an important part of the wealth of their citizens.”

— John Maynard Keynes

This chapter is your *operating manual* for how currencies are actually exchanged in real markets: who trades, why they trade, how trades are quoted and settled, and what kinds of transactions dominate volume.

The conceptual points of the chapter (FX is just Foreign Exchange):

1. **FX exists to move purchasing power across borders.** That’s the *economic* reason the market exists.
2. **FX is also a market microstructure.** Institutions, technology, and market conventions determine *how* prices are formed and trades are executed.
3. **FX quotes are a language.** If you can read that language fluently (direct/indirect, European/American, bid/ask, forwards in points), you can translate any transaction into a cash flow and a risk exposure.

Chapter Roadmap

Major sections:

1. **Functions of the FX market** — what the market *does*.
2. **Structure and participants** — who trades and how the market has evolved.
3. **Transactions** — spot, forwards, swaps (and NDFs), plus why swaps dominate volume.
4. **Rates and quotations** — how to read quotes, compute cross-rates, and spot arbitrage.
5. **Summary** — a compact “mental model” you can keep.

Learning Objectives

After this lecture, you should be able to:

- Explain the three classic functions of the FX market (purchasing power transfer, trade credit, risk hedging).
 - Describe the modern market structure (electronic trading, reduced interdealer/customer separation, prime brokerage, retail aggregation).
 - Distinguish spot, forward, and swap transactions and explain why swaps dominate market turnover.
 - Read currency quotations in multiple conventions (ISO codes, base/quote, European vs. American terms, direct vs. indirect) and interpret bid/ask spreads.
 - Compute cross rates and identify when triangular (intermarket) arbitrage may exist.
 - Interpret forward points/pips and translate them into forward premiums/discounts.
-

1.1 Functions of the Foreign Exchange Market

1.1.1 The three functions (and why they matter)

The foreign exchange (FX) market is the mechanism by which participants:

1. **Transfer purchasing power between countries.**

When a U.S. firm buys from a Japanese supplier, *someone* must end up holding yen and *someone* must end up holding fewer dollars. The FX market is how we convert one money into another so the transaction can clear.

2. **Obtain or provide credit for international trade transactions.**

Goods move slowly relative to money. Inventory in transit has to be financed (letters of credit, bankers' acceptances, etc.). FX markets sit next to (and often inside) the trade finance machinery.

3. **Hedge and transfer exchange-rate risk.**

Anytime you have a future cash flow in a foreign currency, you have a price risk: the home-currency value of that foreign cash flow can change before you receive/pay it. FX markets provide instruments to *transfer* that risk to someone willing to bear it.

Note: international trade forces you to choose *one* invoice currency; that choice allocates FX risk between the parties. The FX market is what allows either side to manage (or speculate on) that risk.

1.1.2 Common misconceptions

- “The FX market is a physical place.”
It’s overwhelmingly an electronic, global, OTC network.
- “FX is only for speculators.”
The core volume exists because of trade, investment, and financing needs; speculation is *layered on top*.
- “Hedging eliminates risk for free.”
Hedging transfers risk and typically locks in a price; it can reduce variance but it also removes the upside.

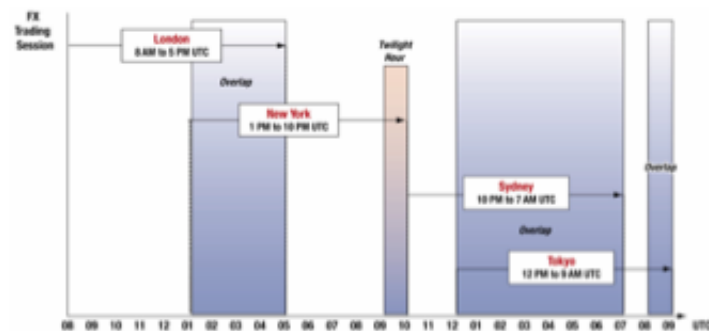
1.2 Structure of the Foreign Exchange Market

1.2.1 The global trading day

FX is a **24-hour market** because major trading centers open and close sequentially.
You can see this better in the slides or text.

Exhibit 5.1

Global Currency Trading: The Trading Day



The currency trading day extends 24 hours per day. The busiest time of the day, which historically was the London and New York overlap, has now started shifting farther East to Asia.



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Figure 1: Global Currency Trading: The Trading Day

Observation: liquidity is not uniform through the day. The market is usually deepest when major centers overlap (historically London–New York; increasingly Asia’s role has grown).

1.2.2 Market participants – the players

A simple and useful classification is:

- **Liquidity seekers:** participants who need FX to do something else (trade, investment, tourism, corporate treasury).
- **Profit seekers:** participants who trade FX *as the business* (dealers, some hedge funds, arbitrageurs/speculators).

1.2.2.1 Bank and nonbank dealers

Dealers act as **market makers**:

- They quote **bid** (what they will pay to buy) and **ask** (what they will charge to sell).
- Their profit is primarily the **bid–ask spread**, adjusted for inventory risk and volatility.

Microstructure logic: a dealer is always managing two risks:

1. **Inventory risk** (holding an open position while prices move)
2. **Adverse selection risk** (the counterparty may know more than you)

1.2.2.2 Commercial and investment transactors

Importers/exporters, multinational corporations, portfolio investors, tourists.

- FX is *incidental*—it is necessary to execute a commercial or investment transaction, but not the purpose.

1.2.2.3 Speculators and arbitrageurs

- **Speculators** take positions to profit from expected exchange-rate changes. (Significantly more risk.)
- **Arbitrageurs** exploit price inconsistencies across markets (simultaneous buy/sell to lock in profit).

A clean distinction:

- Speculation: profit from **directional movement**.
- Arbitrage: profit from **relative mispricing** with (near) no net exposure.

1.2.2.4 Central banks and treasuries

Central banks and treasuries enter FX markets to:

- Manage **foreign exchange reserves**
- Conduct **foreign exchange intervention** (influence the value of their currency through supply and demand)

Critical difference: they are not primarily profit motivated and may act as *willing loss takers* if it serves national objectives.

1.2.2.5 Foreign exchange brokers

Brokers match counterparties:

- They facilitate trades **without becoming principals** in the transaction.
- They earn commissions/fees rather than spreads.

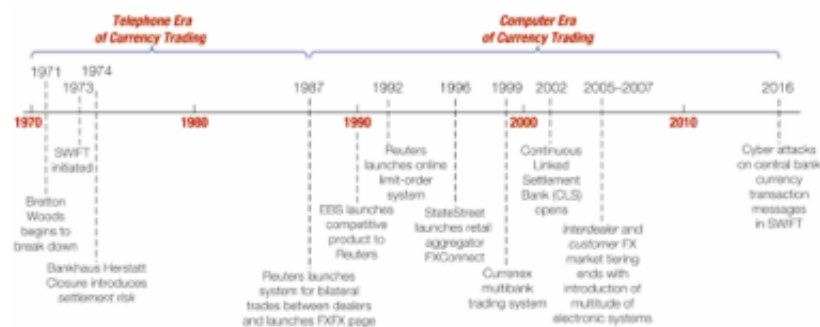
1.2.3 Evolution of the market

The collapse of Bretton Woods and the move toward floating rates expanded the market dramatically. Trading evolved by improving three components:

1. **Trading** (how orders are executed)
2. **Messaging** (how instructions are communicated)
3. **Settlement** (how currencies are actually delivered)

Exhibit 5.2

Evolution of the Modern Currency Trading Marketplace



All foreign currency transactions require some combination of three elements: 1) trading, 2) messaging, and 3) settlement. The modern history of currency trading involves the evolution of these elements from voice-based and paper-based communication to the use of computers for trading, financial messaging, and multilateral netting settlement.



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Figure 2: Evolution of the Modern Currency Trading Marketplace

1.2.4 The evolution of FX trading

1.2.4.1 1985: the telephone era

- Voice-based trading.
- Limited transparency; dealers had less information about “the market” at any moment.
- Higher operational error risk (paper-based confirmations).

1.2.4.2 1990s: computers + better information

- Electronic messaging improves speed and reduces errors.
- Dealers still “knew” who they were trading with, but market data improved.

1.2.4.3 2010 – today: electronic dominance and tiering breakdown

Key developments:

- **Multi-bank trading systems (MBT)** and **single-bank trading systems (SBT)**
- **Prime brokerage** enables smaller players to access liquidity
- **Retail aggregators** connect small customers into larger pools

Conceptual payoff: technology compresses spreads and increases access—but it can also change the form (not the existence) of market manipulation.

1.2.5 The three components of FX trades

Every FX trade has three separable pieces:

1. **The trade agreement** (price, amount, value date)
2. **Payment/settlement instructions** (messaging)
3. **Final settlement** (actual delivery of currencies)

This separation matters because *operational and settlement risk* can exist even when the price risk is hedged.

1.2.6 FX market manipulation: “Fixing the fix”

Benchmark rates (like a daily “fix”) can be important for valuation and index tracking. Allegations (and enforcement actions) around 2013–2014 highlighted how:

- Concentrated trading windows can create incentives to influence a benchmark.
- Electronic trading reduces some manipulation channels but can enable new ones.

1.2.7 The FX Global Code of Conduct (2016)

Following misconduct revelations, global principles emphasized:

- Ethics
- Governance
- Information sharing
- Execution
- Risk management and compliance
- Confirmation and settlement processes

1.3 Transactions in the Foreign Exchange Market

The FX market is commonly described by three transaction categories:

1. **Spot**
2. **Forward**
3. **Swap** (including variations like NDFs)

1.3.1 Spot transactions

A **spot transaction** is an agreement to exchange currencies with delivery typically on **T+2** (two business days after the trade date, for most major currency pairs).

Use-cases:

- An exporter receives foreign currency today and wants home currency.
- An importer needs foreign currency immediately to make payment.

1.3.2 Forward transactions

An **outright forward** is an agreement today to exchange currencies at a **specified rate** on a **future value date**.

Core idea: a forward rate converts an uncertain future spot price into a known price *today*.

- **Hedging**: lock in the home-currency value of a foreign-currency cash flow.
- **Speculation**: take a view on where spot will be at maturity.

Language nuance:

“Buying euros forward” and “selling dollars forward” describe the same trade—just from opposite currency perspectives.

1.3.3 Swap transactions

A **swap** is the simultaneous purchase and sale of a currency for two different value dates (same counterparty).

Swaps dominate turnover largely because they are used to:

- Manage short-term liquidity and funding needs
- Roll positions and hedge exposures efficiently
- Transform maturities (e.g., borrowing/lending implicitly in a foreign currency)

1.3.3.1 Spot against forward

This is the classic structure: buy (or sell) spot and do the opposite forward.

Interpretation: it resembles **fully collateralized borrowing/lending** in FX form.

1.3.3.2 Forward-forward swaps

Both legs are forward (two future value dates). Used for term funding and hedging.

1.3.3.3 Nondeliverable forwards (NDFs)

An **NDF** is a forward contract that settles in a hard currency (often USD) **without delivering** the restricted currency.

Typical when:

- The currency has capital controls or limited convertibility.
- Market participants still need hedging or price discovery offshore.

Teaching anchor: NDFs are an institutional workaround—“we’ll settle the P&L in dollars rather than deliver the restricted currency.”

1.3.4 Settlement timing and value dates

Appendix 5

Long Description for Exhibit 5.5

The details are as follows: The delivery periods mentioned are today, tomorrow, the day after tomorrow, two or more days after tomorrow, and later than two or more days after tomorrow. The following list provides the delivery period for each category. Spot transactions stretch from tomorrow to the day after tomorrow. Overnight transactions stretch from today to the day after tomorrow. Forward or Outright Forward deliveries may occur two or more days after tomorrow. Swap transactions include spot against forward, forward, forward swaps, and nondeliverable forwards. Swap transactions can take place from the day after tomorrow to any period later than two or more days after tomorrow.

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Figure 3: FX Transactions and Settlement

Vocabulary:

- **Value date:** the settlement date when currency is delivered.

- **Overnight:** extremely short-dated trades (often to manage liquidity).

1.3.5 Size of the FX market

The FX market is the world's largest financial market by turnover.

Appendix 6

Long Description for Exhibit 5.6

The x-axis shows years from 1989 to 2019 in increments of 3 years. The y-axis shows U.S. dollars in billions from 0 to 7000 in increments of 1000. Each stacked bar is divided into segments representing spot transactions, outright forwards, FX swaps, and options and other. The details of the graph showing daily turnover and percentage change respectively are as follows: 1989: 590, blank; 1992: 820, plus 39; 1995: 1190, plus 45; 1998: 1527, plus 28; 2001: 1239, minus 19; 2004: 1934, plus 54; 2007: 3324, plus 72; 2010: 3973, plus 20; 2013: 5357, plus 35; 2016: 5066, minus 5; and 2019: 6590, plus 30.

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Figure 4: Global FX Market Turnover, 1989–2019

Conceptual takeaway: if swaps are the largest category, the FX market is not “mostly about tourists” or “mostly about importing/exporting goods.” It’s deeply tied to global financing.

1.3.6 Geographical distribution

Appendix 7

Long Description for Exhibit 5.7

The x-axis shows years from 2013 to 2019 in increments of 3. The y-axis shows U.S. dollars in billions from 0 to 4000 in increments of 500. The F X trading percentage for the selected countries in 2013, 2016, and 2019 respectively are as follows: United Kingdom: 41, 36, 53; United States: 19, 19, 20; Singapore: 6, 8, 9; Hong Kong: 4, 7, 9; Japan: 6, 6, 6; Switzerland: 3, 2, 4; France: 3, 3, 2; Germany: 2, 2, 2; Australia: 3, 2, 2; and China: 1, 1, 2.

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Figure 5: Top 10 Geographic Trading Centers

Classroom point: London's role as a global hub reflects time zone overlap, institutional depth, and historical network effects.

1.3.7 Currency composition

Appendix 8 (1 of 4)

Long Description for Exhibit 5.8

The first table lists trading for different currency pairs versus the dollar for the years 2001, 2004, 2007, 2010, 2013, 2016, and 2019. The details respectively are as follows: USD, EUR; Euro; 2001, 30.0; 2004, 28.0; 2007, 26.8; 2010, 27.7; 2013, 24.1; 2016, 23.1; and 2019, 24.0. USD, JPY; Japanese yen; 2001, 20.2; 2004, 17.0; 2007, 13.2; 2010, 14.3; 2013, 18.3; 2016, 17.8; and 2019, 13.2. USD, GBP; British pound; 2001, 10.4; 2004, 13.4; 2007, 11.6; 2010, 9.1; 2013, 8.8; 2016, 9.3; and 2019, 9.6. USD, AUD; Australian dollar; 2001, 4.1; 2004, 5.5; 2007, 5.6; 2010, 6.3; 2013, 6.8; 2016, 5.2; and 2019, 5.4. USD, CAD; Canadian dollar; 2001, 4.3; 2004, 4; 2007, 3.8; 2010, 4.6; 2013, 3.7; 2016, 4.3; and 2019, 4.4. USD, CNY; Chinese yuan; 2001, No data; 2004, No data; 2007, No data; 2010, 0.8; 2013, 2.1; 2016, 3.8; and 2019, 4.1. USD, CHF; Swiss franc; 2001, 4.8; 2004, 4.3; 2007, 4.5; 2010, 4.2; 2013, 3.4; 2016, 3.6; and 2019, 3.5. USD, HKD; Hong Kong dollar; 2001, No data; 2004, No data; 2007, No data; 2010, 2.1; 2013, 1.3; 2016, 1.5; and 2019, 3.3. USD, KRW; Korean won; 2001, No data; 2004, No data; 2007, No data; 2010, 1.5; 2013, 1.1; 2016, 1.5; and 2019, 1.9. USD, INR; Indian rupee; 2001, No data; 2004, No data; 2007, No data; 2010, 0.9; 2013, 0.9; 2016, 1.1; and 2019, 1.7. USD, SGD; Singapore dollar; 2001, No data; 2004, No data; 2007, No data; 2010, No data; 2013, 1.2; 2016, 1.6; and 2019, 1.7. USD, NZD; New Zealand dollar; 2001, No data; 2004,



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Figure 6: Daily FX Trading by Currency Pair

The U.S. dollar appears on one side of most trades. That does **not** mean the U.S. must be a counterparty—it means the dollar is often the *vehicle currency* used to move between two less-liquid currencies.

1.3.8 Likely student misconceptions

- **“Spot means instantaneous.”**
In interbank markets it typically settles T+2.
- **“Forward contracts are like futures.”**
Both are forward-looking, but forwards are OTC and customized; futures are standardized and exchange-traded.
- **“Swaps are rare/advanced.”**
They are the *largest* volume category because funding and liquidity management is constant.

1.3.9 Check for understanding

1. A U.S. firm will pay JPY in 3 months. What FX instrument most directly locks in the USD cost today?

2. Why might a bank use swaps repeatedly rather than a single long-dated forward?
-

1.4 Foreign Exchange Rates and Quotations

1.4.1 Currency symbols and ISO codes

Currencies may be referenced by:

- ISO codes (USD, GBP, JPY, EUR) are standard for electronic trading.
- **ISO 4217** three-letter codes (USD, EUR, JPY, GBP, etc.)

In institutional electronic trading, ISO codes dominate.

1.4.2 Exchange rate quotes: base and quote currency

Every quote is written as:

$$\text{CUR}_1/\text{CUR}_2$$

- **Base (unit) currency:** CUR_1 (the “1 unit” currency)
- **Quote (price) currency:** CUR_2 (how many units you pay to get 1 unit of base)

Example: **EUR/USD 1.2174** means 1 euro costs 1.2174 dollars.

1.4.3 Market conventions

Different conventions can describe the same economic price.

1.4.3.1 European terms

“Foreign currency per 1 USD” (USD is the base).

Example: USD/JPY 130.00 means USD 1 = JPY 130.

1.4.3.2 American terms

“USD per 1 unit of foreign currency” (USD is the quote).

Example: EUR/USD 1.10 means EUR 1 = USD 1.10.

1.4.3.3 Currency nicknames

You will see informal terms in markets and news (e.g., “cable” for GBP/USD, “kiwi” for NZD). The key is to translate the nickname back to a base/quote pair.

1.4.3.4 Direct and indirect quotations

These depend on who is “home”:

- **Direct quote:** home currency price of 1 unit of foreign currency.
- **Indirect quote:** foreign currency price of 1 unit of home currency.

For a U.S. resident, USD/EUR is direct; EUR/USD is indirect.

1.4.3.5 Bid and ask rates

Interbank quotes are two-sided:

- **Bid:** dealer buys the base currency (you sell it to the dealer)
- **Ask (offer):** dealer sells the base currency (you buy it from the dealer)

Appendix 9

Long Description for Exhibit 5.10

The details are as follows: base currency, EUR; quote currency, USD; you can sell 1 euro for 1.2170 dollars in “Bid”; you can buy 1 euro for 1.2178 dollars in “Ask”; or traders may quote only the last two digits on a rate as 1.2170 or 78.

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Figure 7: Bid, Ask, and Mid-Point Quotation

A useful operational rule:

- *You buy at the ask; you sell at the bid.*

Appendix 10

Long Description for Exhibit 5.11

The details for currency, symbol, code, USD equivalent, and currency per USD respectively are as follows: Americas: Argentina, peso, Ps, A R S; 0.0107; and 93.8528. Brazil, real; R dollar; BR L; 0.1913; 5.2287. Canada, dollar, C dollar; CAD; 0.8261; 1.2105. Chile, peso; dollar; CLP; 0.001435; 697. Mexico, new peso; dollars; MX N; 0.0502; 19.9392. Asia: Australia, dollar; A dollar; AU D; 0.7829; 1.2773. China, yuan; Y; CN Y; 0.1558; 6.4166. Hong Kong, dollar; H K dollar; HK G; 0.1288; 7.7656. India, rupees; Rs; IN R; 0.01361; 73.48555. Indonesia, rupiah; Rp; I D R; 0.0000704; 14198. Japan, yen; Y; JP Y; 0.00919; 108.84; Singapore, dollar; S dollar; SG D; 0.7543; 1.3257. South Korea, won; W; KR W; 0.0008954; 1116.83. Thailand, baht; B; TH B; 0.03213; 31.12. Vietnam, dong; d; VND; 0.00004336; 23062. Europe: Czech Republic, koruna; Kc; CZ K; 0.04744; 21.08. Denmark, korne; Dkr; DK K; 0.1631; 6.1307. Euro area, euro; e; EU R; 1.213; 0.8244. Norway, krone; Nkr; NO K; 0.1209; 8.2727. Russia, ruble; R; R U B; 0.01345; 74.327. Sweden, krona; SKr; SE K; 0.1198; 0.9011. Switzerland, franc; Fr.; CH F; 1.1098; 0.9011. Middle East and Africa: Egypt, pound; pound symbol; E G P; 0.0638; 15.6713. Israel, shekel; Shk; IL S; 0.3068; 3.2593. Saudi Arabia, riyal; S R; SAR; 0.2667; 3.7502. South Africa, rand; R; ZAR; 0.0712; 14.0492.

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Figure 8: Exchange Rates: New York Closing Snapshot

Why do spreads exist? processing costs, inventory costs, volatility, and market competition/liquidity.

1.4.4 Cross rates

Many currency pairs are thinly traded. Their rate is often implied via a third currency (commonly USD).

Example logic:

If you know USD/EUR and USD/GBP, then

$$\text{EUR/GBP} = \frac{\text{USD/GBP}}{\text{USD/EUR}}$$

(Keep track of units like you would in dimensional analysis.)

1.4.5 Intermarket (triangular) arbitrage

Arbitrage exists when the implied cross-rate differs from the quoted cross-rate enough to overcome transaction costs.

Appendix 11

Long Description for Exhibit 5.12

The banks at the three edges of the triangle are Citibank New York, Barclays Bank, London; and Dresdner Bank. the steps are as follows: Step 1. Trader sells USD 1,000,000 to Barclays Bank at USD 1.5585 equals GBP 1.00. Step 2. Trader receives GBP 641,643. Step 3. Trader sells GBP 641,643 to Dresdner Bank at EUR 1.1722 equals GBP 1.00. Step 4. Trader receives EUR 752,133 from Dresdner Bank. Step 5. Trader sells EUR 752,133 to Citibank at USD 1.3297 equals EUR 1.00. Step 6. Trader receives USD 1,000,112.

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Figure 9: Triangular Arbitrage by a Market Trader

A safe teaching sequence:

1. Start with one currency (e.g., USD).
2. Convert to a second currency at the dealer's bid/ask.
3. Convert to a third currency.
4. Convert back to the original.
5. If you end with more than you started (after costs), an arbitrage exists.

1.4.6 Forward quotations

Forward rates are often quoted as **forward points** (or pips) relative to spot.

- Spot is quoted “outright” (all digits).
- Forwards are quoted as the *adjustment* to spot.

Exhibit 5.13

Spot and Forward Quotations for the Euro and Japanese Yen

Spot and Forward Quotations for the Euro and Japanese yen								
Euro: Spot and Forward (\$=€1.00)					Japanese yen: Spot and Forward (¥=\$1.00)			
Term	Bid		Ask		Bid		Ask	
	Points	Rate	Points	Rate	Points	Rate	Points	Rate
Cash rates								
Spot		1.0897		1.0901		118.27		118.37
1 week	3	1.0900	4	1.0905	-10	118.17	-9	118.28
1 month	17	1.0914	19	1.0920	-51	117.76	-50	117.87
2 months	35	1.0932	36	1.0937	-95	117.32	-93	117.44
3 months	53	1.0950	54	1.0955	-143	116.84	-140	116.97
4 months	72	1.0969	76	1.0977	-195	116.32	-190	116.47
5 months	90	1.0987	95	1.0996	-240	115.87	-237	116.00
6 months	112	1.1009	113	1.1014	-288	115.39	-287	115.50
9 months	175	1.1072	177	1.1078	-435	113.92	-429	114.08
1 year	242	1.1139	245	1.1146	-584	112.43	-581	112.56
Swap rates								
2 years	481	1.1378	522	1.1423	-1150	106.77	-1129	107.08
3 years	750	1.1647	810	1.1711	-1748	100.79	-1698	101.39
4 years	960	1.1857	1039	1.1940	-2185	96.42	-2115	97.22
5 years	1129	1.2026	1276	1.2177	-2592	92.35	-2490	93.47

[For long description, see slide 62: Appendix 12](#)



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Figure 10: Spot and Forward Quotations for the Euro and Japanese Yen

1.4.6.1 Forward premium/discount (annualized)

A common calculation uses mid-rates:

$$\text{Forward premium (annualized)} \approx \left(\frac{F - S}{S} \right) \times \frac{360}{n}$$

- If ($F > S$): the foreign currency is at a **forward premium** (in American terms).
- If ($F < S$): it is at a **forward discount**.

Important teaching note: the sign and interpretation can flip depending on quote convention (American vs European terms). The safest approach is: define your quote clearly, compute $((F-S)/S)$, and interpret the meaning in words (“more or fewer USD per 1 unit of foreign currency”).

1.4.7 Likely student misconceptions

- “**Direct vs. indirect is the same as American vs. European.**”
They are related but not identical; direct/indirect depends on who is “home.”
- “**Bid/ask doesn’t matter in calculations.**”
It matters *a lot* for arbitrage and any realistic conversion path.

- **“Cross rates are just multiplying numbers.”**
Cross rates are unit algebra; always track currency units.

1.4.8 Check for understanding

1. If EUR/USD is 1.10, what is USD/EUR?
2. If a dealer quotes EUR/USD 1.1000–1.1004, what price do you pay to buy euros?
3. If the implied EUR/GBP differs from the quoted EUR/GBP, what additional information do you need before declaring “free money”?

1.5 Summary

A compact way to remember Chapter 5:

- **Purpose:** FX moves purchasing power, provides trade finance mechanisms, and enables risk transfer.
- **Structure:** a global OTC network with dealers as market makers; technology has widened access and changed microstructure.
- **Transactions:** spot (near-term delivery), forwards (future delivery at fixed rate), swaps (two legs; dominant volume), and NDFs (offshore settlement in hard currency).
- **Language:** quotes have conventions—base/quote, direct/indirect, European/American terms, bid/ask.
- **Computation:** cross rates and arbitrage are unit-consistent bookkeeping; forward points translate spot into forward.

If you leave with one skill, make it this: **given any FX quote, you can state (i) what is being priced, (ii) what you pay to buy and what you receive to sell, and (iii) how that quote maps into a real cash flow and risk exposure.**